

Study Questions: Planning

1. Describe the decision process of choosing your next crochet project using the concept of **expected free energy**. What factors do you weigh, and how do they map onto the Active Inference framework?
2. Compare the expected free energy of three potential projects: a simple dishcloth, a moderately complex hat, and a challenging lace shawl. Which factors push expected free energy up or down for each?
3. When estimating how much yarn a project will require, you are planning under uncertainty. Describe the sources of uncertainty in this estimate and how your generative model manages them.
4. A crocheter always buys one extra skein "just in case." Explain this strategy as a policy for managing irreducible uncertainty in yarn estimation.
5. How does **difficulty assessment** require the generative model to evaluate its own capacity? What does it mean for the crocheter to have an accurate self-model versus an overconfident or underconfident one?
6. When choosing colors for a multi-color project, the crocheter must predict how colors will look together in the finished fabric. Describe this as a visual simulation — running the generative model forward in the aesthetic domain.
7. The concept of a "yarn stash" reduces planning uncertainty by expanding available options. Explain how having a diverse stash changes the crocheter's expected free energy when choosing a new project.
8. A crocheter plans to make five holiday gifts and has three months to complete them. Describe the temporal planning process: what must the generative model predict, and what sources of uncertainty are involved?

9. After completing a large project, some crocheters experience "project choice paralysis" — they cannot decide what to make next. Explain this in Active Inference terms: what happens when too many options produce similar expected free energy?
10. How does the experience of past projects update the crocheter's planning model? Give an example of a lesson learned from a previous project that changes how you plan future ones.
11. A crocheter selects a pattern rated "Advanced" despite being at an intermediate skill level, motivated by the desire to learn new techniques. How does **epistemic value** (the value of learning) factor into the expected free energy calculation?
12. When a crocheter decides to "wing it" — start a project without a pattern, making design decisions as they go — how does planning differ from following a written pattern? What role does real-time model simulation play?
13. Describe the planning differences between a project for yourself and a project for someone else (a gift). How does the recipient's preferences introduce additional uncertainty into the planning model?
14. A crocheter discovers mid-project that they will run out of yarn. Describe the replanning process: what options are available, and how does the crocheter evaluate the expected free energy of each?
15. How does the availability of online pattern databases (like Ravelry) change the planning process? Does access to thousands of patterns increase or decrease planning difficulty?
16. A crocheter is deciding between a quick, satisfying project and a long, ambitious one. Describe the trade-off between short-term and long-term expected free energy in this decision.
17. When planning a garment (like a sweater), the crocheter must account for body measurements, ease, and fit. How does the complexity of the generative model increase compared to planning a flat item like a scarf?
18. Some crocheters keep a "project queue" — a list of future projects ranked by priority. How does maintaining a queue function as a planning tool in Active Inference terms?

19. Describe how a crocheter's planning ability changes over their career. What aspects of planning improve most with experience?
20. Reflect on your own planning process for crochet projects. What are your strengths and weaknesses as a planner, and how might a deeper understanding of Active Inference help you make better project choices?